CUSTOMER SEGMENTATION

Using Data Science & K-Means Clustering

Data Science Internship Project Report

Project Type	Unsupervised Machine Learning
Algorithm	K-Means Clustering
Dataset Size	200 Customers
Number of Segments	5
Primary Features	Annual Income and Spending Score
Technology	Python, Pandas, Scikit-learn, Matplotlib, Seaborn

Prepared for internship evaluation and academic documentation

1. Abstract

Customer segmentation is an important application of data science in marketing and retail analytics. The objective of this project is to identify groups of customers with similar characteristics and spending behavior so that businesses can develop more targeted marketing strategies. A customer dataset containing age, annual income, and spending score information was analyzed using Python. After preprocessing and exploratory data analysis, the K-Means clustering algorithm was applied. The Elbow Method was used to select five clusters. The final model divided 200 customers into five distinct segments, including high-value customers, young high-spending customers, high-income low-spending customers, and lower-spending customer groups. The resulting segments provide actionable insights for personalized promotions, customer retention, loyalty programs, and efficient allocation of marketing resources.

2. Introduction

Businesses serve customers with different needs, purchasing capacity, and engagement levels. Treating all customers in exactly the same way can result in inefficient marketing and missed opportunities. Customer segmentation addresses this challenge by dividing customers into meaningful groups according to similarities in their characteristics or behavior.

2.1 Problem Statement

The project aims to analyze customer data and automatically group customers into distinct segments using an unsupervised machine learning technique. The resulting clusters should be interpretable from a business perspective and useful for targeted marketing decisions.

2.2 Objectives

- Load and inspect the customer dataset.
- Clean and preprocess the data for analysis.
- Perform exploratory data analysis using statistical summaries and visualizations.
- Determine a suitable number of customer groups using the Elbow Method.
- Apply K-Means clustering to create customer segments.
- Visualize and interpret the resulting clusters.
- Generate practical business recommendations from the segmentation.

3. Tools and Technologies

Technology	Purpose
Python	Core programming and analysis
Pandas	Data loading, cleaning, transformation, and aggregation
Scikit-learn	Standardization and K-Means clustering
Matplotlib	Data visualization
Seaborn	Statistical and correlation visualizations
Jupyter Notebook	Interactive analysis and documentation

4. Dataset Description

The analysis uses a 200-customer dataset containing customer identifiers and demographic/behavioral attributes. The key numerical variables used for segmentation are Age, Annual Income (in thousands of dollars), and Spending Score (1–100). Annual income represents purchasing capacity, while the spending score represents customer purchasing behavior or engagement.

Feature	Description	Role
CustomerID	Unique customer identifier	Identifier
Gender	Customer gender	Descriptive
Age	Customer age in years	Analysis feature
Annual Income (k$)	Annual income in thousands of dollars	Clustering feature
Spending Score (1–100)	Customer spending/engagement score	Clustering feature

5. Data Preprocessing

- Loaded the customer dataset using Pandas.
- Inspected data types, missing values, duplicate records, and numerical ranges.
- Selected relevant numerical features for clustering.
- Standardized clustering variables so that differences in scale did not unfairly influence K-Means distance calculations.
- Prepared the cleaned feature matrix for unsupervised learning.

6. Exploratory Data Analysis

Exploratory analysis was performed before clustering to understand the distribution of customer age, income, and spending behavior. The visualizations also helped identify the structure of the customer population.

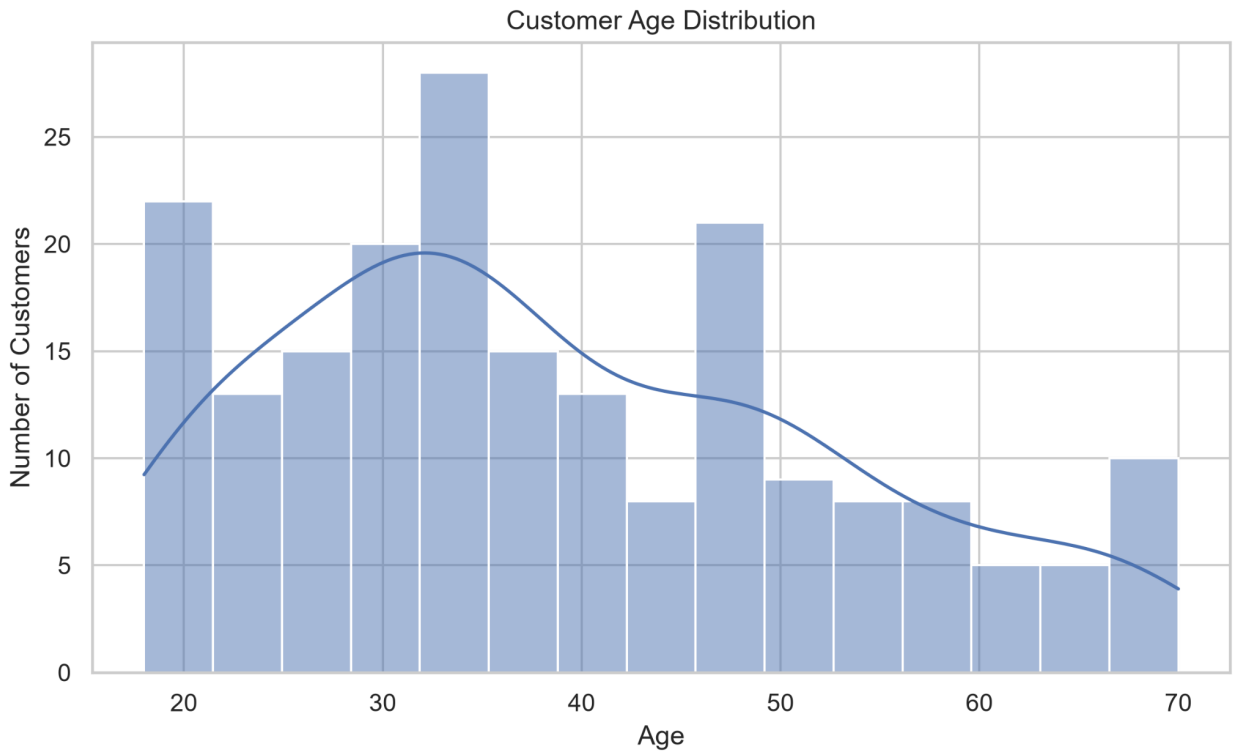


Figure 1. Customer age distribution.

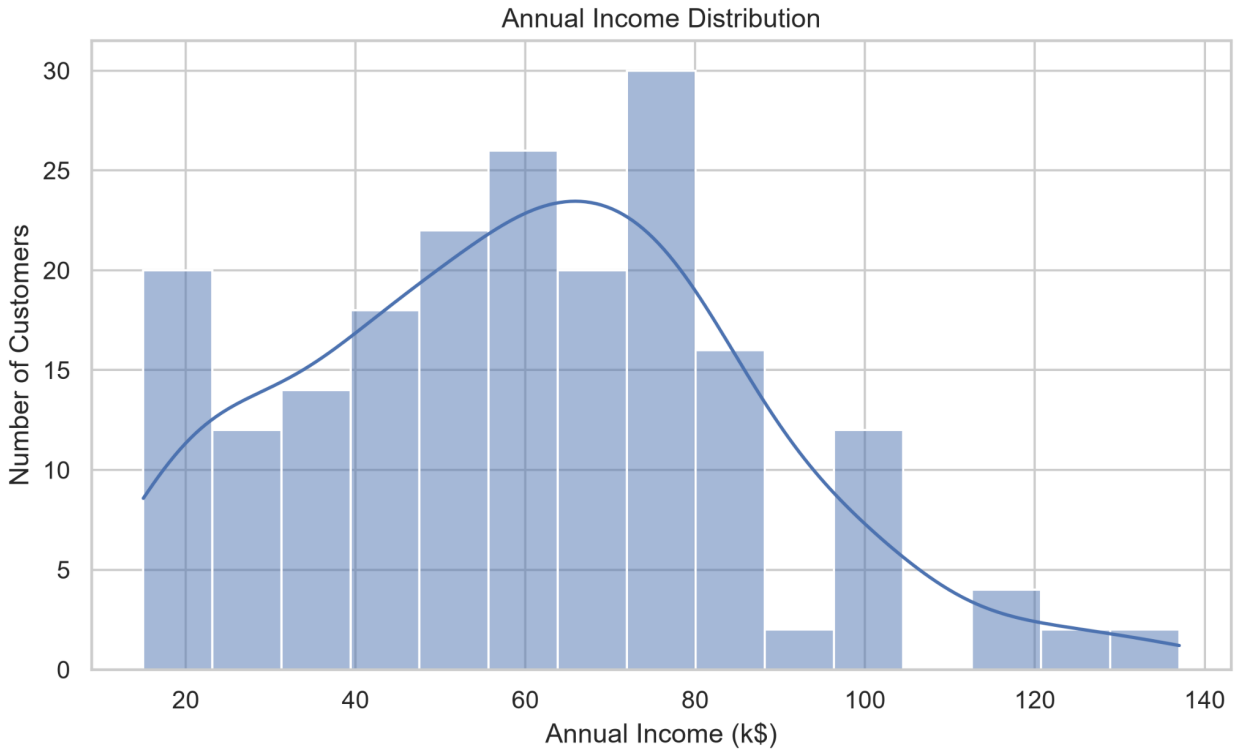


Figure 2. Annual income distribution.

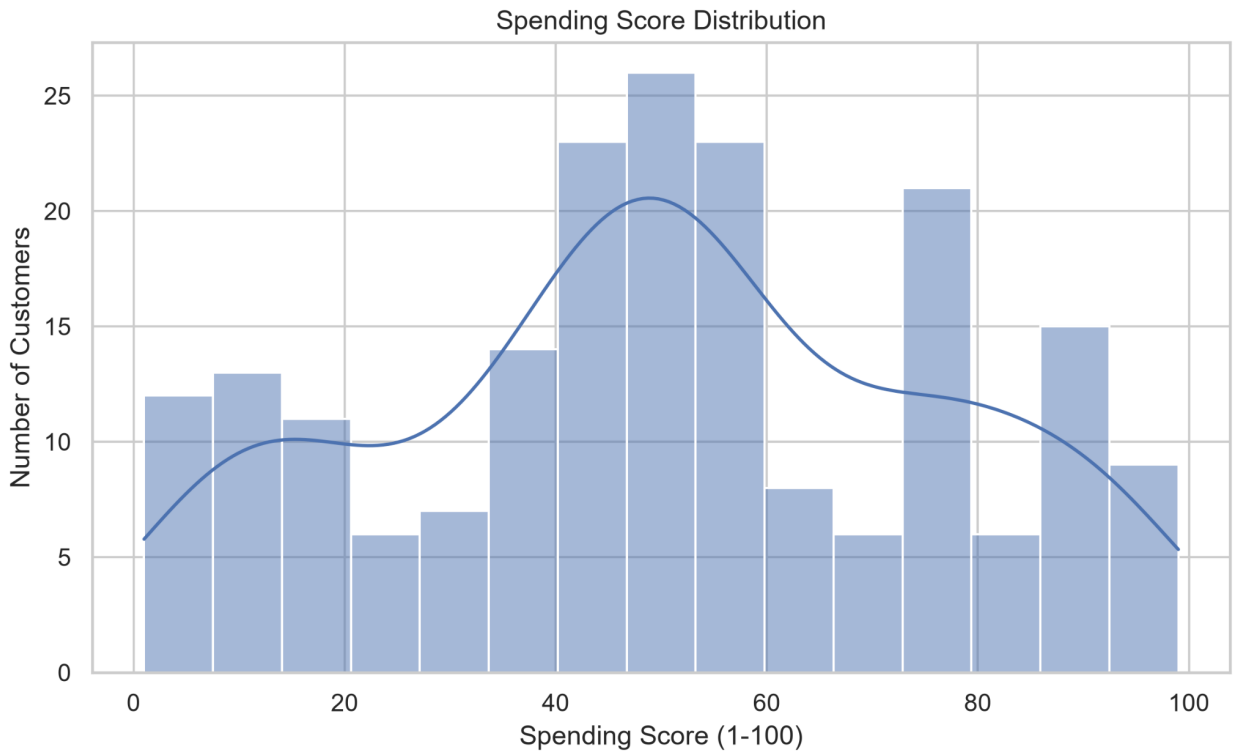


Figure 3. Spending score distribution.

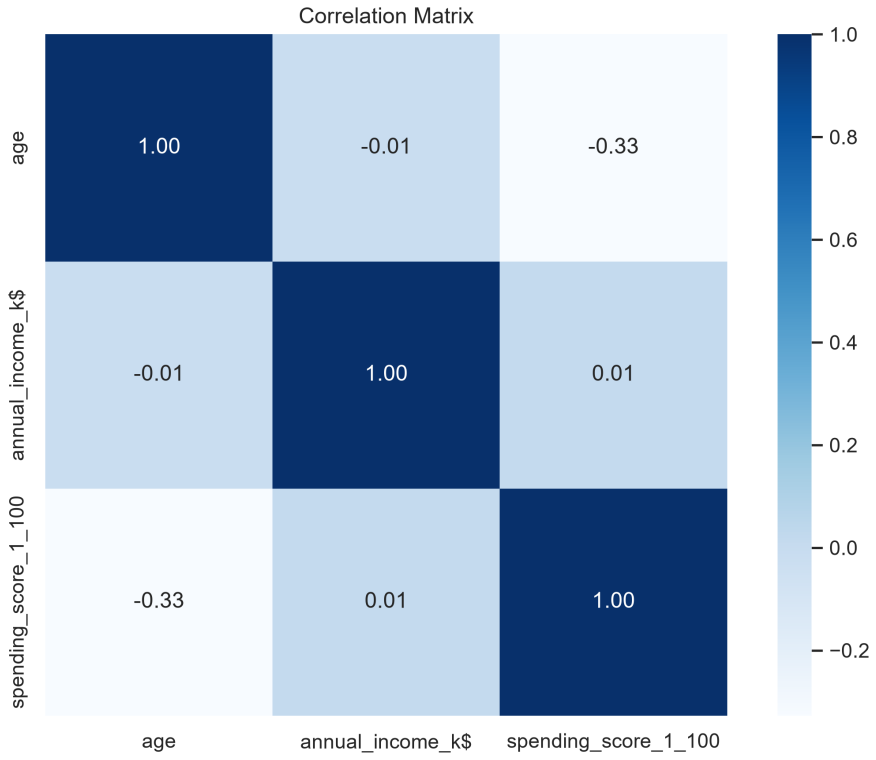


Figure 4. Correlation matrix of numerical variables.

The correlation matrix shows a weak relationship between annual income and spending score (approximately 0.01). This indicates that income alone is not a reliable indicator of spending behavior. Age and spending score show a moderate negative relationship (approximately -0.33).

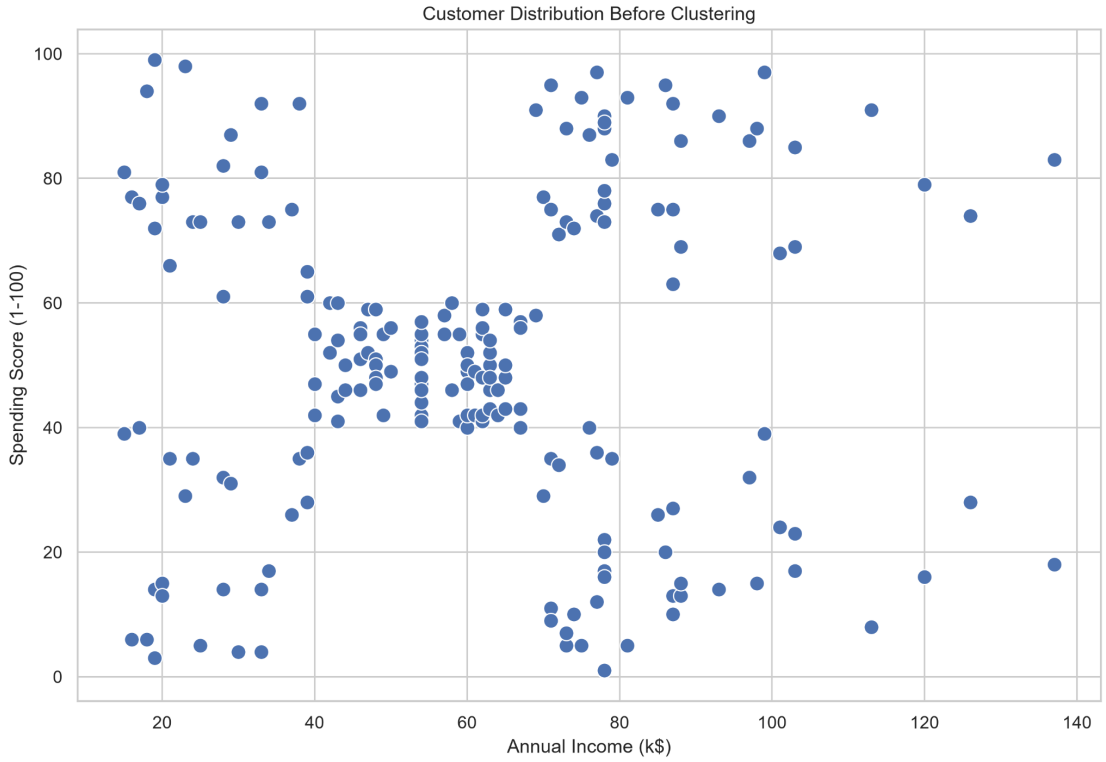


Figure 5. Customer distribution before clustering.

7. K-Means Clustering Methodology

K-Means is an unsupervised learning algorithm that partitions observations into K groups by minimizing the within-cluster sum of squared distances between observations and their assigned cluster centers. In this project, standardized customer features were used to reduce scale-related bias.

7.1 Selecting the Number of Clusters

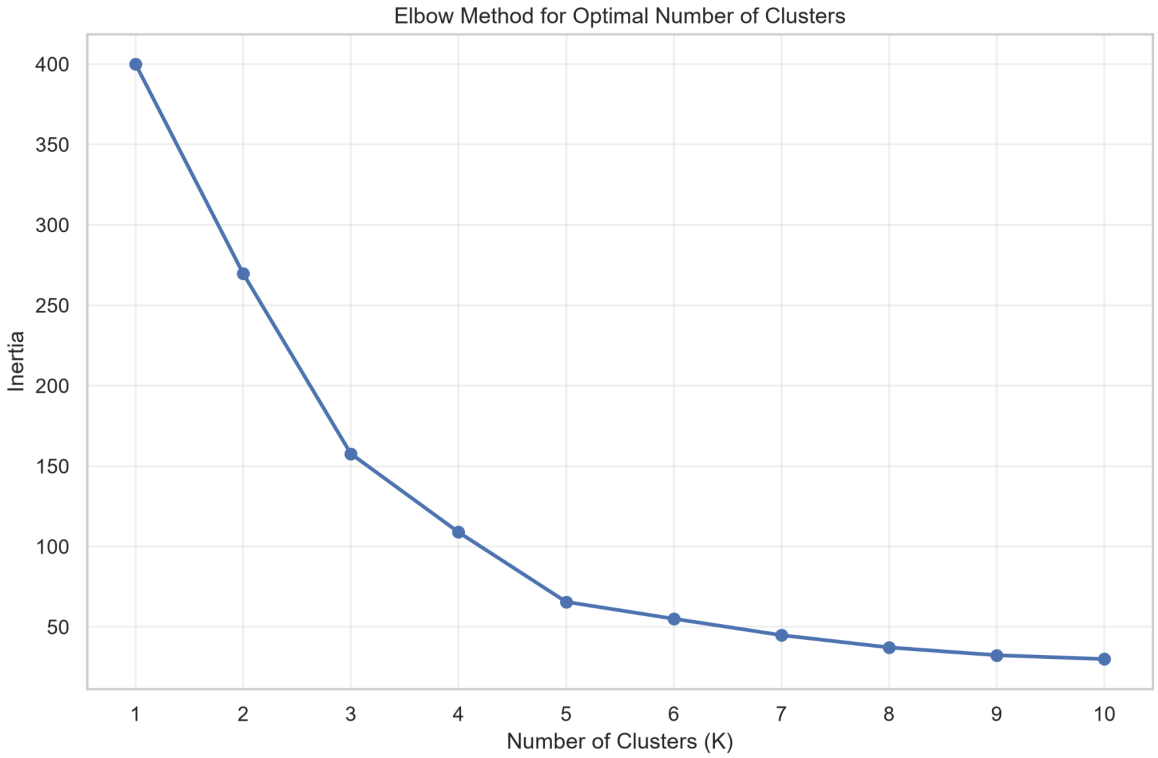


Figure 6. Elbow Method for selecting K.

The Elbow Method evaluates the model's inertia for different values of K. The graph shows a large reduction in inertia through K=5, after which the improvement becomes considerably smaller. Therefore, K=5 was selected as the practical number of customer segments.

8. Customer Segmentation Results

The final K-Means model produced five customer segments from 200 observations. The following table contains the actual cluster statistics generated by the analysis.

Cluster	Customers	%	Avg Age	Avg Income	Avg Spending	Segment
0	81	40.5%	42.72	$55.30K	49.52	Low/Mod. Income Low-Spending Customers
1	39	19.5%	32.69	$86.54K	82.13	High-Value Customers
2	22	11.0%	25.27	$25.73K	79.36	Low/Mod. Income High-Spending Customers
3	35	17.5%	41.11	$88.20K	17.11	High-Income Low-Spending Customers
4	23	11.5%	45.22	$26.30K	20.91	Low/Mod. Income Low-Spending Customers

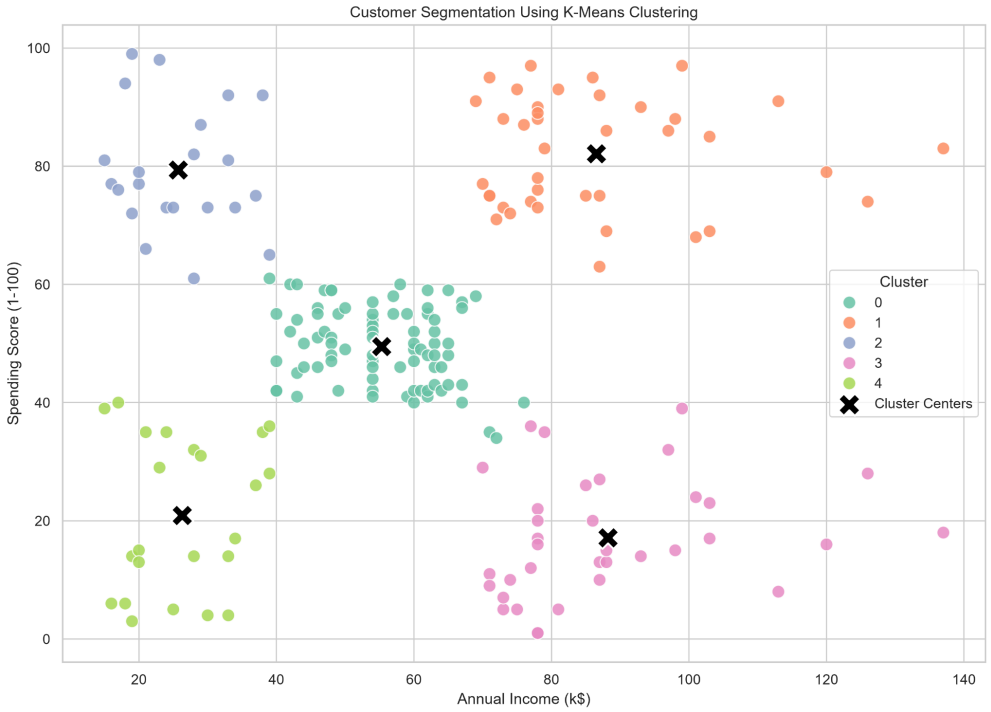


Figure 7. Customer segments based on annual income and spending score.

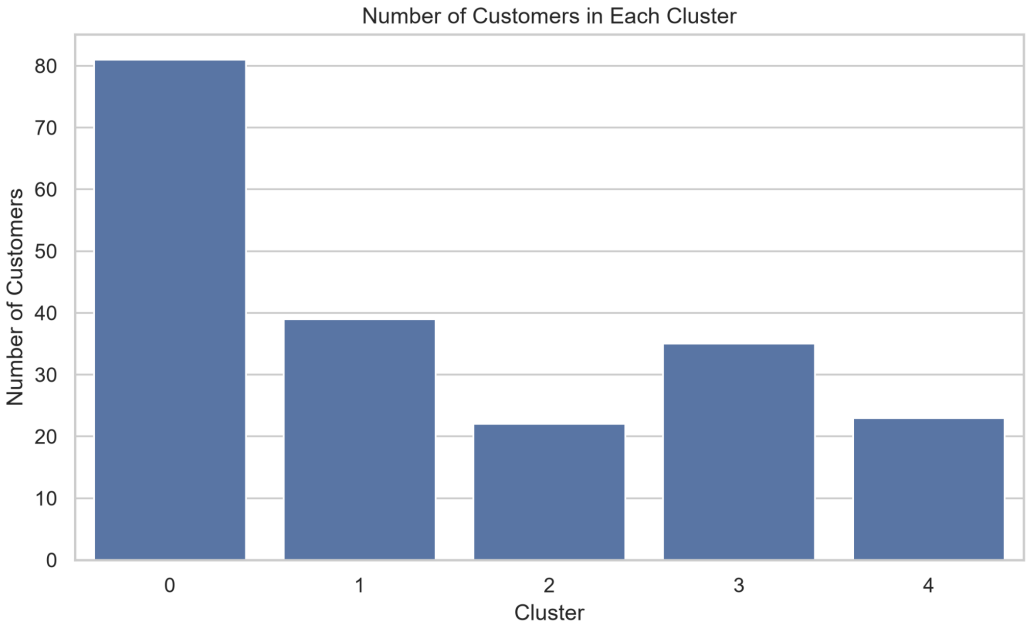


Figure 8. Number of customers in each cluster.

9. Business Insights and Interpretation

Cluster 0 — Mainstream / Moderate-Spending Customers

This is the largest group with 81 customers (40.5%). The average income is $55.30K and the average spending score is 49.52. Businesses can target this broad segment with general promotions, cross-selling, loyalty incentives, and personalized recommendations.

Cluster 1 — High-Value Customers

This group contains 39 customers (19.5%) with an average income of $86.54K and a high spending score of 82.13. It is the strongest premium segment. Recommended actions include loyalty programs, exclusive offers, premium products, personalized recommendations, and early access campaigns.

Cluster 2 — Young High-Spending Customers

This segment contains 22 customers (11.0%) with an average age of 25.27 years, income of $25.73K, and spending score of 79.36. Businesses can use youth-oriented campaigns, social-media marketing, referral programs, affordable premium products, and limited-time promotions.

Cluster 3 — High-Income Low-Spending Customers

This segment contains 35 customers (17.5%). Despite having the highest average income of $88.20K, its spending score is only 17.11. This group represents an important re-engagement opportunity. Personalized marketing, customer feedback, product demonstrations, and targeted incentives can be used to understand and improve their engagement.

Cluster 4 — Low-Income Low-Spending Customers

This group contains 23 customers (11.5%) with average income of $26.30K and spending score of 20.91. Value bundles, budget-friendly products, seasonal discounts, and cost-efficient campaigns are appropriate strategies for this segment.

10. Key Findings

- The dataset contains 200 customers and was successfully segmented into five clusters.
- Cluster 0 is the largest customer group, representing 40.5% of the dataset.
- Cluster 1 is the high-value segment with high income and high spending behavior.
- Cluster 3 demonstrates that high income does not necessarily result in high spending.
- Cluster 2 represents younger customers with relatively low income but high spending scores.
- The near-zero income-to-spending correlation supports the value of behavioral segmentation.
- The Elbow Method supports the selection of five clusters for the final K-Means model.

11. Conclusion

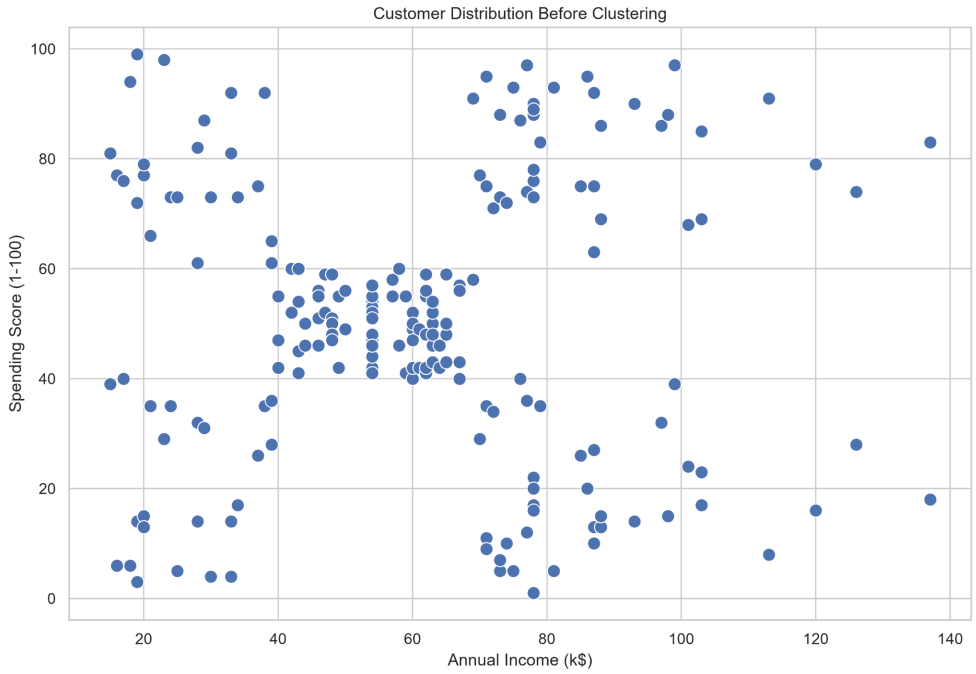
The Customer Segmentation project successfully demonstrates how data science and unsupervised machine learning can be applied to a practical marketing problem. K-Means clustering identified five distinct customer groups with substantially different combinations of income and spending behavior. The segmentation provides a structured basis for differentiated marketing strategies and demonstrates that customer value cannot be determined from income alone. The project also provides an end-to-end workflow covering data preprocessing, exploratory analysis, model selection, clustering, visualization, and business interpretation.

12. Future Scope

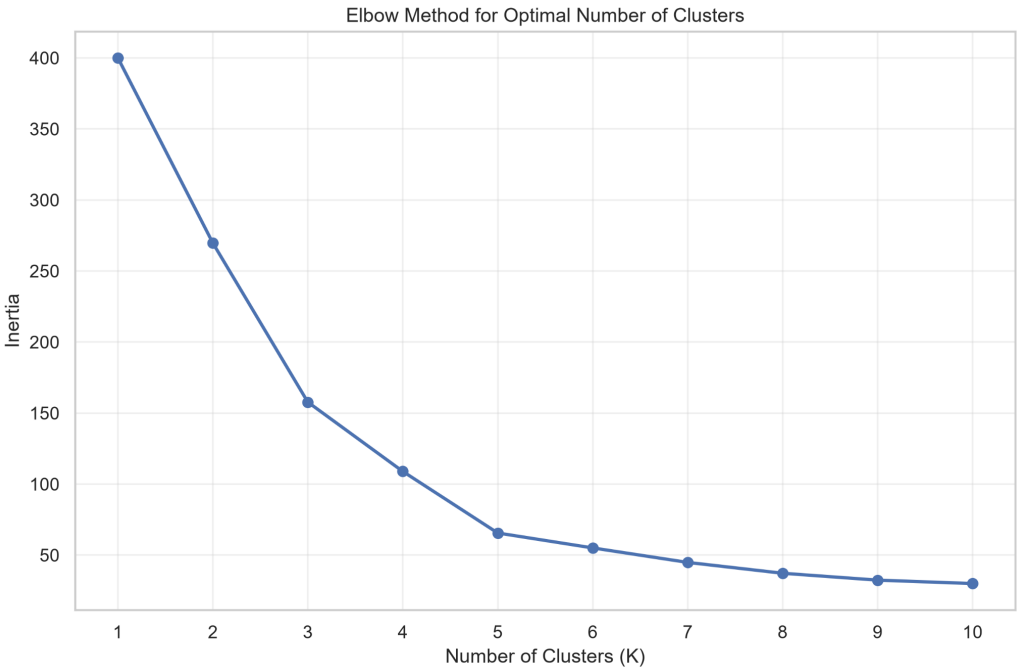
- Add purchase frequency, transaction value, recency, and product-category information.
- Use RFM analysis together with K-Means for richer customer profiling.
- Compare K-Means with hierarchical clustering or DBSCAN.
- Build an interactive Streamlit dashboard for business users.
- Track customer segments over time to identify movement between segments.
- Integrate campaign-response data to evaluate the effectiveness of segment-specific marketing.

13. Project Screenshots

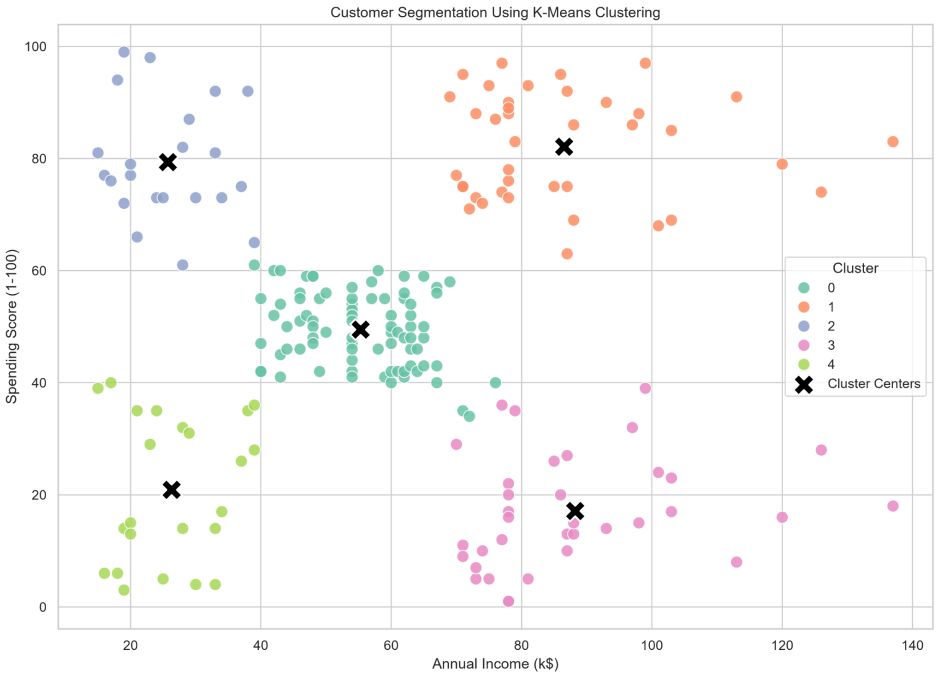
The following outputs were generated during the project and can be used as evidence for internship evaluation.



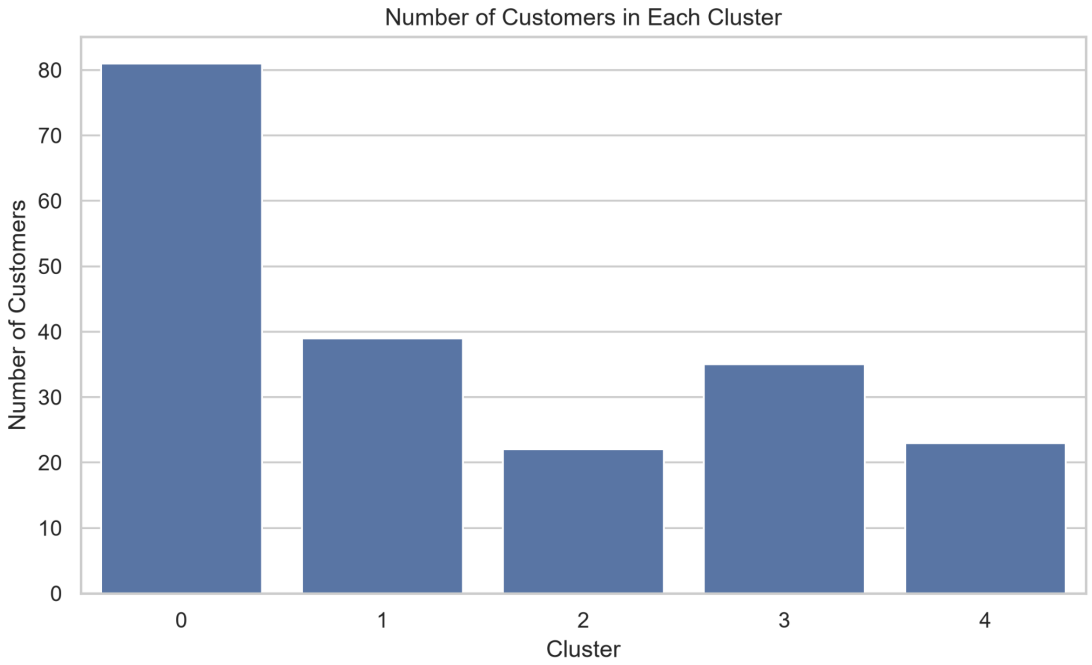
Screenshot A — Customer distribution before clustering



Screenshot B — Elbow Method



Screenshot C — Final cluster visualization



Screenshot D — Cluster distribution

14. References

- Scikit-learn documentation — K-Means clustering and preprocessing.
- Pandas documentation — data loading, cleaning, and analysis.
- Matplotlib documentation — data visualization.
- Seaborn documentation — statistical visualization.